

```
import pandas as pd
```

```
data = pd.read_csv('data/training.csv')
```

```
data.isna().sum().sum()
```

```
np.int64(0)
```

```
metrics = pd.DataFrame(columns=["best_params_Ca", "best_params_P", "best_params_pH",  
"best_params_SOC", "best_params_Sand", "model", "Ca", "P", "pH", "SOC", "Sand", "MCRMSE"])
```

U datasetu nema null vrednosti.

```
data.select_dtypes(include='str')
```

	PIDN	Depth
0	XNhoFZW5	Topsoil
1	9XNspFTd	Subsoil
2	WDId41qG	Topsoil
3	JrrJf1mN	Subsoil
4	ZoIitegA	Topsoil
...	...	...
1152	bdcNNrbi	Topsoil
1153	6HBVKZwh	Subsoil
1154	5eLY5nw7	Topsoil
1155	gsSGXhX6	Subsoil
1156	A6IpBbfZ	Topsoil

1157 rows × 2 columns

Izbacujemo PDIN koji predstavlja identifikator uzorka

```
data.drop('PIDN', inplace=True, axis=1)
```

Pretvaramo Depth obelezje u numericke vrednosti

```
data['Depth'] = data['Depth'].replace({'Topsoil' : 0 , 'Subsoil' : 1})
```

Inicijalizacija X i Y za MultiOutputRegressor

```
Y = data.loc[:, ['Ca', 'P', 'pH', 'SOC', 'Sand']]
X = data.drop(['Ca', 'P', 'pH', 'SOC', 'Sand'], axis=1)
Y.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1157 entries, 0 to 1156
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Ca          1157 non-null   float64
1   P           1157 non-null   float64
2   pH          1157 non-null   float64
3   SOC         1157 non-null   float64
4   Sand        1157 non-null   float64
dtypes: float64(5)
memory usage: 45.3 KB
```

Provera korelacija

```
import numpy as np

cor_mat = data.corr()
cor_mat_ut = cor_mat.where(np.triu(np.ones(cor_mat.shape), k=1).astype(np.bool))
cols = cor_mat_ut.columns
correlated_columns = []
for column in cols:
    if any(cor_mat_ut[column] > 0.95):
        correlated_columns.append(column)
correlated_columns
```

```
data_bez_korelacija = data.drop(correlated_columns, axis=1)
```

data_bez_korelacija

	m7497.96	BSAN	BSAV	CTI	ELEV	EVI	LSTD	LSTN	REF2	
0	0.302553	-0.630435	-0.783875	-0.364146	1.165479	1.062682	-0.716713	-0.090016	-0.537106	-0.
1	0.270192	-0.630435	-0.783875	-0.364146	1.165479	1.062682	-0.716713	-0.090016	-0.537106	-0.
2	0.317433	-0.753623	-0.929451	-0.633972	1.544098	1.156705	-1.282552	-0.088336	-0.631725	-0.
3	0.261116	-0.753623	-0.929451	-0.633972	1.544098	1.156705	-1.282552	-0.088336	-0.631725	-0.
4	0.260038	-0.688406	-0.884658	-0.583576	1.276837	1.191691	-1.206971	0.011420	-0.528757	-0.
...	...	...	...	...	...	...	...	...	...	...
1152	0.207530	-0.724638	-0.772676	0.038610	1.501782	0.306851	-0.511141	-0.481023	-0.940631	-0.
1153	0.182376	-0.724638	-0.772676	0.038610	1.501782	0.306851	-0.511141	-0.481023	-0.940631	-0.
1154	0.146829	-0.695652	-0.795073	-0.639966	1.287973	0.659621	-0.549317	-0.646528	-0.515770	-0.
1155	0.091910	-0.695652	-0.795073	-0.639966	1.287973	0.659621	-0.549317	-0.646528	-0.515770	-0.
1156	0.471388	-0.797101	-0.783875	0.087648	1.265702	0.271137	-0.245935	-0.444938	-0.914657	-0.

1157 rows × 19 columns

```
type(data_bez_korelacija['Depth'])
```

pandas.Series

X i Y za MultipleOutputRegressor za bez korelacija

```
Y_bez_korelacija = data_bez_korelacija.loc[:,['Ca', 'P', 'pH', 'SOC', 'Sand']]
X_bez_korelacija = data_bez_korelacija.drop(['Ca', 'P', 'pH', 'SOC', 'Sand'], axis=1)
X_bez_korelacija.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1157 entries, 0 to 1156
Data columns (total 14 columns):
#   Column      Non-Null Count  Dtype
---  -
0   m7497.96    1157 non-null    float64
1   BSAN        1157 non-null    float64
2   BSAV        1157 non-null    float64
3   CTI         1157 non-null    float64
4   ELEV        1157 non-null    float64
5   EVI         1157 non-null    float64
6   LSTD        1157 non-null    float64
7   LSTN        1157 non-null    float64
8   REF2        1157 non-null    float64
9   REF3        1157 non-null    float64
10  RELI        1157 non-null    float64
11  TMAP        1157 non-null    float64
12  TMFI        1157 non-null    float64
13  Depth       1157 non-null    object
dtypes: float64(13), object(1)
memory usage: 126.7+ KB
```

racunanje korelacije mi je ubilo operativni sistem

Outlier-i

```
from scipy.stats import normaltest

norm = []
columns = data.select_dtypes(include='number').columns.tolist()
for column in columns:
    p = normaltest(data[column])
    if p.pvalue < 0.05:
        norm.append(False)
    else:
        norm.append(True)

print(pd.DataFrame(norm).sum(), data.shape)
```

```
0    418
dtype: int64 (1157, 3599)
```

Od 3599 kolona svega 418 ima normalnu raspodelu, pa ne mozemo koristiti z_score za uklanjanje outlier-a

```

data_numeric = data.select_dtypes(include='number')

data_bez_outliera = data

q1 = data_numeric.quantile(0.25)
q3 = data_numeric.quantile(0.75)
iqr = q3 - q1

columns = data_numeric.columns
for column in columns:
    if (column == 'Ca') | (column == 'P') | (column == 'pH') | (column == 'SOC') |
(column == 'Sand'):
        continue
    data_bez_outliera = data_bez_outliera[(data_bez_outliera[column] >= q1[column] - 1.5
* iqr[column]) & (data_bez_outliera[column] <= q3[column] + 1.5 * iqr[column])]

```

```
data_bez_outliera.shape
```

```
(715, 3599)
```

Izbacivanjem outliera izgubili smo preveliku kolicinu podataka, pa cemo ih clippovati

```

data_numeric = data.select_dtypes(include='number')

data_clipped = data

q1 = data_numeric.quantile(0.25)
q3 = data_numeric.quantile(0.75)
iqr = q3 - q1

columns = data_numeric.columns
for column in columns:
    if (column == 'Ca') | (column == 'P') | (column == 'pH') | (column == 'SOC') |
(column == 'Sand'):
        continue
    min = q1[column] - 1.5 * iqr[column]
    max = q3[column] + 1.5 * iqr[column]
    data_clipped[column] = data_clipped[column].clip(upper=max, lower=min)

```

X i Y za MultipleOutputRegressor za clipped podatke

```
Y_clipped = data_clipped.loc[:,['Ca', 'P', 'pH', 'SOC', 'Sand']]
X_clipped = data_clipped.drop(['Ca', 'P', 'pH', 'SOC', 'Sand'], axis=1)
X_clipped.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1157 entries, 0 to 1156
Columns: 3594 entries, m7497.96 to Depth
dtypes: float64(3593), object(1)
memory usage: 31.7+ MB
```

Modeli

Linearna regresija

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import GridSearchCV
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler

model_lin_reg = LinearRegression(n_jobs=-1)
pipeline_lin_reg = Pipeline([
    ('scaler', StandardScaler()),
    ('model_lin_reg', model_lin_reg),
])
grid_params = {
    'scaler' : [None, StandardScaler()],
    'model_lin_reg__fit_intercept' : [False, True],
}

grid_lin_reg = GridSearchCV(pipeline_lin_reg, grid_params, n_jobs=-1, cv=5, scoring=
'neg_mean_squared_error')
```

Regresiono stablo

```

from sklearn.tree import DecisionTreeRegressor

model_tree = DecisionTreeRegressor(random_state=7)
pipeline_tree = Pipeline([
    ('model_tree', model_tree),
])

parameters_tree = {
    "model_tree__ccp_alpha" : [0.0, 0.05, 0.0025],
    "model_tree__max_depth" : [2, 5, 7, 9, 11],
    "model_tree__criterion" : ['squared_error', 'friedman_mse', 'absolute_error'],
}

grid_tree = GridSearchCV(pipeline_tree, parameters_tree, n_jobs=-1, cv=5, scoring=
    'neg_mean_squared_error')

```

KNN

```

from sklearn.neighbors import KNeighborsRegressor

knn = KNeighborsRegressor()

knn_pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ("knn", knn),
])

parameters_knn = {
    "knn__n_neighbors" : [3, 5, 7, 9],
    "knn__weights" : ['uniform', 'distance'],
    "knn__metric" : ['minkowski'],
    "knn__algorithm" : [ 'kd_tree', 'brute']
}

grid_knn = GridSearchCV(knn_pipeline, parameters_knn, n_jobs=-1 , cv=5, scoring=
    "neg_mean_squared_error")

```

Random Forest

```

from sklearn.ensemble import RandomForestRegressor

forest = RandomForestRegressor( n_estimators=100)

random_forest_pipeline = Pipeline([
    ("random_forest", forest),
])

parameters_random_forest = {
    "random_forest__n_estimators" : [50, 100, 150, 200],
    "random_forest__criterion" : ['squared_error', 'absolute_error', 'friedman_mse'],
    "random_forest__max_depth" : [2, 5, 7, 9]
}

grid_random_forest = GridSearchCV(random_forest_pipeline, parameters_random_forest,
n_jobs=-1 , cv=5, scoring="neg_mean_squared_error")

```

XG boost

```

import xgboost as xgb

xg_boost_pipeline = Pipeline ([
    ("model_xg_boost", xgb.XGBRegressor(eval_metric='rmse',device='cuda'))
])

parameters_xg_boost = {
    "model_xg_boost__max_depth":    [4, 5, 6],
    "model_xg_boost__n_estimators": [140, 150, 160],
    "model_xg_boost__learning_rate": [0.01, 0.015]}

grid_xg_boost = GridSearchCV(xg_boost_pipeline, parameters_xg_boost, n_jobs=-1 , cv=5,
scoring="neg_mean_squared_error")

```

Sa outlierima

Izdvajamo izlazna obelezja

```

from sklearn.model_selection import train_test_split
from sklearn.metrics import root_mean_squared_error

def the_funkcija(grid, trening_set, validacija_set, output_name):
    metrics={}
    # izdvajanje izlaznih obelezja
    y_Ca = trening_set['Ca']
    y_P  = trening_set['P']
    y_pH = trening_set['pH']
    y_SOC = trening_set['SOC']
    y_Sand = trening_set['Sand']

```



```

X = trening_set.drop(['Ca', 'P', 'pH', 'SOC', 'Sand'], axis=1)
# train test split
X_train_Ca, X_test_Ca, y_train_Ca, y_test_Ca = train_test_split(X, y_Ca, test_size=0.2,
, random_state=7)
X_train_P, X_test_P, y_train_P, y_test_P = train_test_split(X, y_P, test_size=0.2,
random_state=7)
X_train_pH, X_test_pH, y_train_pH, y_test_pH = train_test_split(X, y_pH, test_size=0.2,
, random_state=7)
X_train_SOC, X_test_SOC, y_train_SOC, y_test_SOC = train_test_split(X, y_SOC,
test_size=0.2, random_state=7)
X_train_Sand, X_test_Sand, y_train_Sand, y_test_Sand = train_test_split(X, y_Sand,
test_size=0.2, random_state=7)
# krosvalidacija
grid_Ca = grid.fit(X_train_Ca, y_train_Ca)
best_model_Ca = grid_Ca.best_estimator_
best_params_Ca = grid_Ca.best_params_

metrics['best_params_Ca'] = best_params_Ca

grid_P = grid.fit(X_train_P, y_train_P)
best_model_P = grid_P.best_estimator_
best_params_P = grid_P.best_params_

metrics['best_params_P'] = best_params_P

grid_pH = grid.fit(X_train_pH, y_train_pH)
best_model_pH = grid_pH.best_estimator_
best_params_pH = grid_pH.best_params_

metrics['best_params_pH'] = best_params_pH

grid_SOC = grid.fit(X_train_SOC, y_train_SOC)
best_model_SOC = grid_SOC.best_estimator_
best_params_SOC = grid_SOC.best_params_

metrics['best_params_SOC'] = best_params_SOC

grid_Sand = grid.fit(X_train_Sand, y_train_Sand)
best_model_Sand = grid_Sand.best_estimator_
best_params_Sand = grid_Sand.best_params_

metrics['best_params_Sand'] = best_params_Sand

# kalkulacija rmse
y_RMSE_Ca = root_mean_squared_error(best_model_Ca.predict(X_test_Ca), y_test_Ca)
y_RMSE_P = root_mean_squared_error(best_model_P.predict(X_test_P), y_test_P)
y_RMSE_pH = root_mean_squared_error(best_model_pH.predict(X_test_pH), y_test_pH)
y_RMSE_SOC = root_mean_squared_error(best_model_SOC.predict(X_test_SOC), y_test_SOC)
y_RMSE_Sand = root_mean_squared_error(best_model_Sand.predict(X_test_Sand),
y_test_Sand)

# validacija
prediction = pd.DataFrame()

```

```

prediction['PIDN'] = validacija_set['PIDN']
validacija_set.drop('PIDN', inplace=True, axis=1)

prediction['Ca'] = pd.DataFrame(best_model_Ca.set_params(**best_params_Ca).fit(X,
y_Ca).predict(validacija_set))
prediction['P'] = pd.DataFrame(best_model_P.set_params(**best_params_P).fit(X, y_P).
predict(validacija_set))
prediction['pH'] = pd.DataFrame(best_model_pH.set_params(**best_params_pH).fit(X,
y_pH).predict(validacija_set))
prediction['SOC'] = pd.DataFrame(best_model_SOC.set_params(**best_params_SOC).fit(X,
y_SOC).predict(validacija_set))
prediction['Sand'] = pd.DataFrame(best_model_Sand.set_params(**best_params_Sand).fit
(X, y_Sand).predict(validacija_set))

prediction.to_csv("predictions/" + output_name + ".csv", index=False)

metrics['model'] = output_name
metrics['Ca'] = y_RMSE_Ca
metrics['P'] = y_RMSE_P
metrics['pH'] = y_RMSE_pH
metrics['SOC'] = y_RMSE_SOC
metrics['Sand'] = y_RMSE_Sand
metrics['MCRMSE'] = (y_RMSE_Ca + y_RMSE_P + y_RMSE_pH + y_RMSE_SOC + y_RMSE_Sand)/5

return metrics

```

Linearna regresija inicijalizacija

```

sorted_test = pd.read_csv('data/sorted_test.csv')
test_lin_reg = sorted_test.copy()
test_lin_reg['Depth'] = test_lin_reg['Depth'].replace({'Topsoil' : 0 , 'Subsoil' : 1})

```

```

row = the_funkcija(grid_lin_reg, data, test_lin_reg, "lin_reg")
print(row)
metrics=pd.concat([metrics, pd.DataFrame([row])])

```

```

{'best_params_Ca': {'model_lin_reg__fit_intercept': True, 'scaler': None},
'best_params_P': {'model_lin_reg__fit_intercept': False, 'scaler': None},
'best_params_pH': {'model_lin_reg__fit_intercept': False, 'scaler': StandardScaler()},
'best_params_SOC': {'model_lin_reg__fit_intercept': False, 'scaler': None},
'best_params_Sand': {'model_lin_reg__fit_intercept': False, 'scaler': None}, 'model':
'lin_reg', 'Ca': 0.5049645752975458, 'P': 1.6576700943521183, 'pH': 0.7543211407602292,
'SOC': 0.7051158437573091, 'Sand': 0.7294245874734764, 'MCRMSE': 0.8702992483281358}

```

Regresiono stablo inicijalizacija

```

test_tree = sorted_test.copy()
test_tree['Depth'] = test_tree['Depth'].replace({'Topsoil' : 0 , 'Subsoil' : 1})

```

```

row = the_funkcija(grid_tree, data, test_tree, "tree")
print(row)
metrics=pd.concat([metrics, pd.DataFrame([row])])

{'best_params_Ca': {'model_tree__ccp_alpha': 0.0025, 'model_tree__criterion':
'squared_error', 'model_tree__max_depth': 11}, 'best_params_P': {'model_tree__ccp_alpha':
0.05, 'model_tree__criterion': 'friedman_mse', 'model_tree__max_depth': 9},
'best_params_pH': {'model_tree__ccp_alpha': 0.0025, 'model_tree__criterion':
'absolute_error', 'model_tree__max_depth': 7}, 'best_params_SOC':
{'model_tree__ccp_alpha': 0.0025, 'model_tree__criterion': 'absolute_error',
'model_tree__max_depth': 7}, 'best_params_Sand': {'model_tree__ccp_alpha': 0.0025,
'model_tree__criterion': 'squared_error', 'model_tree__max_depth': 11}, 'model': 'tree',
'Ca': 0.5648514501553454, 'P': 1.0588995128583363, 'pH': 0.6745977967977185, 'SOC':
0.5302018230051582, 'Sand': 0.5241819155477588, 'MCRMSE': 0.6705464996728635}

```

KNN inicijalizacija

```

test_knn = sorted_test.copy()
test_knn['Depth'] = test_knn['Depth'].replace({'Topsoil' : 0 , 'Subsoil' : 1})

```

```

row = the_funkcija(grid_knn, data, test_knn, "knn")
print(row)
metrics=pd.concat([metrics, pd.DataFrame([row])])

{'best_params_Ca': {'knn__algorithm': 'kd_tree', 'knn__metric': 'minkowski',
'knn__n_neighbors': 3, 'knn__weights': 'distance'}, 'best_params_P': {'knn__algorithm':
'brute', 'knn__metric': 'minkowski', 'knn__n_neighbors': 5, 'knn__weights': 'distance'},
'best_params_pH': {'knn__algorithm': 'brute', 'knn__metric': 'minkowski',
'knn__n_neighbors': 3, 'knn__weights': 'distance'}, 'best_params_SOC': {'knn__algorithm':
'kd_tree', 'knn__metric': 'minkowski', 'knn__n_neighbors': 3, 'knn__weights':
'distance'}, 'best_params_Sand': {'knn__algorithm': 'brute', 'knn__metric': 'minkowski',
'knn__n_neighbors': 7, 'knn__weights': 'distance'}, 'model': 'knn', 'Ca':
0.3707050255092884, 'P': 0.7548587025994591, 'pH': 0.5371621638791397, 'SOC':
0.4639591210947606, 'Sand': 0.4528866776324533, 'MCRMSE': 0.5159143381430202}

```

Bez outliera

```

test_lin_reg_c = sorted_test.copy()
test_lin_reg_c['Depth'] = test_lin_reg_c['Depth'].replace({'Topsoil' : 0 , 'Subsoil' : 1})

```

```

row = the_funkcija(grid_lin_reg, data_clipped, test_lin_reg_c, "lin_reg_clipped")
print(row)
metrics=pd.concat([metrics, pd.DataFrame([row])])

{'best_params_Ca': {'model_lin_reg__fit_intercept': True, 'scaler': None},
'best_params_P': {'model_lin_reg__fit_intercept': False, 'scaler': None},
'best_params_pH': {'model_lin_reg__fit_intercept': False, 'scaler': StandardScaler()},
'best_params_SOC': {'model_lin_reg__fit_intercept': False, 'scaler': None},
'best_params_Sand': {'model_lin_reg__fit_intercept': False, 'scaler': None}, 'model':
'lin_reg_clipped', 'Ca': 0.5049645752975458, 'P': 1.6576700943521183, 'pH':
0.7543211407602292, 'SOC': 0.7051158437573091, 'Sand': 0.7294245874734764, 'MCRMSE':
0.8702992483281358}

```

Zaključujemo da linearna regresija trenirana na clippovanim podacima radi isto

```

test_tree_clipped = sorted_test.copy()
test_tree_clipped['Depth'] = test_tree_clipped['Depth'].replace({'Topsoil' : 0 , 'Subsoil'
: 1})

```

```

row = the_funkcija(grid_tree, data_clipped, test_tree_clipped, "tree_clipped")
print(row)
metrics=pd.concat([metrics, pd.DataFrame([row])])

{'best_params_Ca': {'model_tree__ccp_alpha': 0.0025, 'model_tree__criterion':
'squared_error', 'model_tree__max_depth': 11}, 'best_params_P': {'model_tree__ccp_alpha':
0.05, 'model_tree__criterion': 'friedman_mse', 'model_tree__max_depth': 9},
'best_params_pH': {'model_tree__ccp_alpha': 0.0025, 'model_tree__criterion':
'absolute_error', 'model_tree__max_depth': 7}, 'best_params_SOC':
{'model_tree__ccp_alpha': 0.0025, 'model_tree__criterion': 'absolute_error',
'model_tree__max_depth': 7}, 'best_params_Sand': {'model_tree__ccp_alpha': 0.0025,
'model_tree__criterion': 'squared_error', 'model_tree__max_depth': 11}, 'model':
'tree_clipped', 'Ca': 0.5648514501553454, 'P': 1.0588995128583363, 'pH':
0.6745977967977185, 'SOC': 0.5302018230051582, 'Sand': 0.5241819155477588, 'MCRMSE':
0.6705464996728635}

```

```

test_knn_clipped = sorted_test.copy()
test_knn_clipped['Depth'] = test_knn_clipped['Depth'].replace({'Topsoil' : 0 , 'Subsoil'
: 1})

```

```

row = the_funkcija(grid_knn, data_clipped, test_knn_clipped, "knn_clipped")
print(row)
metrics=pd.concat([metrics, pd.DataFrame([row])])

{'best_params_Ca': {'knn__algorithm': 'kd_tree', 'knn__metric': 'minkowski',
'knn__n_neighbors': 3, 'knn__weights': 'distance'}, 'best_params_P': {'knn__algorithm':
'brute', 'knn__metric': 'minkowski', 'knn__n_neighbors': 5, 'knn__weights': 'distance'},
'best_params_pH': {'knn__algorithm': 'brute', 'knn__metric': 'minkowski',
'knn__n_neighbors': 3, 'knn__weights': 'distance'}, 'best_params_SOC': {'knn__algorithm':
'kd_tree', 'knn__metric': 'minkowski', 'knn__n_neighbors': 3, 'knn__weights':
'distance'}, 'best_params_Sand': {'knn__algorithm': 'brute', 'knn__metric': 'minkowski',
'knn__n_neighbors': 7, 'knn__weights': 'distance'}, 'model': 'knn_clipped', 'Ca':
0.3707050255092884, 'P': 0.7548587025994591, 'pH': 0.5371621638791397, 'SOC':
0.4639591210947606, 'Sand': 0.4528866776324533, 'MCRMSE': 0.5159143381430202}

```

Bez koreliranih kolona

```

test_lin_reg_bk = sorted_test.copy()
test_lin_reg_bk['Depth'] = test_lin_reg_bk['Depth'].replace({'Topsoil' : 0 , 'Subsoil' : 1
})
test_lin_reg_bk = test_lin_reg_bk.drop(correlated_columns, axis=1)

```

```

row = the_funkcija(grid_lin_reg, data_bez_korelacija, test_lin_reg_bk,
"lin_reg_bez_korelacija")
print(row)
metrics=pd.concat([metrics, pd.DataFrame([row])])

{'best_params_Ca': {'model_lin_reg__fit_intercept': False, 'scaler': None},
'best_params_P': {'model_lin_reg__fit_intercept': False, 'scaler': StandardScaler()},
'best_params_pH': {'model_lin_reg__fit_intercept': False, 'scaler': None},
'best_params_SOC': {'model_lin_reg__fit_intercept': True, 'scaler': None},
'best_params_Sand': {'model_lin_reg__fit_intercept': False, 'scaler': StandardScaler()},
'model': 'lin_reg_bez_korelacija', 'Ca': 0.8329978812820192, 'P': 0.6988197190977719,
'pH': 0.6801143967048152, 'SOC': 0.8315303328421029, 'Sand': 0.7703700255047135,
'MCRMSE': 0.7627664710862845}

```

```

test_tree_bk = sorted_test.copy()
test_tree_bk['Depth'] = test_tree_bk['Depth'].replace({'Topsoil' : 0 , 'Subsoil' : 1})
test_tree_bk = test_tree_bk.drop(correlated_columns, axis=1)

```

```
row = the_funkcija(grid_tree, data_bez_korelacija, test_tree_bk, "tree_bez_korelacija")
print(row)
metrics=pd.concat([metrics, pd.DataFrame([row])])
```

```
{'best_params_Ca': {'model_tree__ccp_alpha': 0.0, 'model_tree__criterion':
'squared_error', 'model_tree__max_depth': 11}, 'best_params_P': {'model_tree__ccp_alpha':
0.05, 'model_tree__criterion': 'absolute_error', 'model_tree__max_depth': 2},
'best_params_pH': {'model_tree__ccp_alpha': 0.0, 'model_tree__criterion':
'absolute_error', 'model_tree__max_depth': 7}, 'best_params_SOC':
{'model_tree__ccp_alpha': 0.0, 'model_tree__criterion': 'squared_error',
'model_tree__max_depth': 5}, 'best_params_Sand': {'model_tree__ccp_alpha': 0.0025,
'model_tree__criterion': 'squared_error', 'model_tree__max_depth': 7}, 'model':
'tree_bez_korelacija', 'Ca': 0.5108288448240199, 'P': 0.7250483854005842, 'pH':
0.6163802595768648, 'SOC': 0.722614972588117, 'Sand': 0.5259174484953628, 'MCRMSE':
0.6201579821769898}
```

```
test_knn_bk = sorted_test.copy()
test_knn_bk['Depth'] = test_knn_bk['Depth'].replace({'Topsoil' : 0 , 'Subsoil' : 1})
test_knn_bk = test_knn_bk.drop(correlated_columns, axis=1)
```

```
row = the_funkcija(grid_knn, data_bez_korelacija, test_knn_bk, "knn_bez_korelacija")
print(row)
metrics=pd.concat([metrics, pd.DataFrame([row])])
```

```
{'best_params_Ca': {'knn__algorithm': 'kd_tree', 'knn__metric': 'minkowski',
'knn__n_neighbors': 7, 'knn__weights': 'distance'}, 'best_params_P': {'knn__algorithm':
'kd_tree', 'knn__metric': 'minkowski', 'knn__n_neighbors': 7, 'knn__weights': 'uniform'},
'best_params_pH': {'knn__algorithm': 'kd_tree', 'knn__metric': 'minkowski',
'knn__n_neighbors': 9, 'knn__weights': 'distance'}, 'best_params_SOC': {'knn__algorithm':
'kd_tree', 'knn__metric': 'minkowski', 'knn__n_neighbors': 9, 'knn__weights':
'distance'}, 'best_params_Sand': {'knn__algorithm': 'kd_tree', 'knn__metric':
'minkowski', 'knn__n_neighbors': 5, 'knn__weights': 'distance'}, 'model':
'knn_bez_korelacija', 'Ca': 0.6794565588462201, 'P': 0.7635625755021704, 'pH':
0.56808776830409, 'SOC': 0.6291466105575492, 'Sand': 0.543697402020544, 'MCRMSE':
0.6367901830461147}
```

```
test_rf_bk = sorted_test.copy()
test_rf_bk['Depth'] = test_rf_bk['Depth'].replace({'Topsoil' : 0 , 'Subsoil' : 1})
test_rf_bk = test_rf_bk.drop(correlated_columns, axis=1)
```

```

row = the_funkcija(grid_random_forest, data_bez_korelacija, test_rf_bk,
"random_forest_bez_korelacija")
print(row)
metrics=pd.concat([metrics, pd.DataFrame([row])])

```

```

{'best_params_Ca': {'random_forest__criterion': 'friedman_mse',
'random_forest__max_depth': 9, 'random_forest__n_estimators': 50}, 'best_params_P':
{'random_forest__criterion': 'squared_error', 'random_forest__max_depth': 9,
'random_forest__n_estimators': 100}, 'best_params_pH': {'random_forest__criterion':
'absolute_error', 'random_forest__max_depth': 9, 'random_forest__n_estimators': 200},
'best_params_SOC': {'random_forest__criterion': 'squared_error',
'random_forest__max_depth': 9, 'random_forest__n_estimators': 200}, 'best_params_Sand':
{'random_forest__criterion': 'friedman_mse', 'random_forest__max_depth': 9,
'random_forest__n_estimators': 150}, 'model': 'random_forest_bez_korelacija', 'Ca':
0.5286413371604731, 'P': 0.9326512486173428, 'pH': 0.5110060939149481, 'SOC':
0.5401998385581174, 'Sand': 0.4167696471089674, 'MCRMSE': 0.5858536330719697}

```

```

test_xg_boost_bk = sorted_test.copy()
test_xg_boost_bk['Depth'] = test_xg_boost_bk['Depth'].replace({'Topsoil' : 0 , 'Subsoil'
: 1})
test_xg_boost_bk = test_xg_boost_bk.drop(correlated_columns, axis=1)

```

```

data_xg_boost_bk = data_bez_korelacija.copy()

data_xg_boost_bk["Depth"] = pd.to_numeric(data_xg_boost_bk["Depth"], errors="coerce")
test_xg_boost_bk["Depth"] = pd.to_numeric(test_xg_boost_bk["Depth"], errors="coerce")

```

```

#print(data_xg_boost_bk.dtypes)
#print(test_xg_boost_bk.dtypes)

```

```

valid_copy_bk = test_xg_boost_bk.copy()
valid_copy_bk["PIDN"] = sorted_test["PIDN"]

```

```

row = the_funkcija(grid_xg_boost, data_xg_boost_bk, valid_copy_bk,
"xg_boost_bez_korelacija")
print(row)
metrics=pd.concat([metrics, pd.DataFrame([row])])

```

```

# print(sorted_test.columns)
# print(test_xg_boost_bk.columns)

```

```

{'best_params_Ca': {'model_xg_boost__learning_rate': 0.015, 'model_xg_boost__max_depth':
6, 'model_xg_boost__n_estimators': 160}, 'best_params_P':
{'model_xg_boost__learning_rate': 0.01, 'model_xg_boost__max_depth': 5,
'model_xg_boost__n_estimators': 150}, 'best_params_pH': {'model_xg_boost__learning_rate':
0.015, 'model_xg_boost__max_depth': 6, 'model_xg_boost__n_estimators': 160},
'best_params_SOC': {'model_xg_boost__learning_rate': 0.015, 'model_xg_boost__max_depth':
5, 'model_xg_boost__n_estimators': 160}, 'best_params_Sand':
{'model_xg_boost__learning_rate': 0.015, 'model_xg_boost__max_depth': 6,
'model_xg_boost__n_estimators': 160}, 'model': 'xg_boost_bez_korelacija', 'Ca':
0.5223930316940733, 'P': 1.0381065811647154, 'pH': 0.5145140904859381, 'SOC':
0.5949350353014508, 'Sand': 0.4311780110421139, 'MCRMSE': 0.6202253499376582}

```

Polynomial features

broj polinoma je prevelik pre izbacivanja koreliranih kolona, pa samo ovde koristimo polynomial features

ostavili smo samo drugi stepen polinoma jer visi stepeni daju isti rezultat

```
from sklearn.preprocessing import PolynomialFeatures

pipeline_lin_reg_polynomial = Pipeline([
    ('scaler', StandardScaler()),
    ('feature_generator', PolynomialFeatures()),
    ('model_lin_reg', model_lin_reg),
])

lin_reg_params_polynomial = {
    'scaler' : [None, StandardScaler()],
    'feature_generator' : [None, PolynomialFeatures(degree=2)],
    'model_lin_reg__fit_intercept' : [False, True],
}

grid_lin_reg_polynomial = GridSearchCV(pipeline_lin_reg_polynomial,
lin_reg_params_polynomial, n_jobs=-1, cv=5, scoring='neg_mean_squared_error')
```

```
test_lin_reg_bk_pol = sorted_test.copy()
test_lin_reg_bk_pol['Depth'] = test_lin_reg_bk_pol['Depth'].replace({'Topsoil' : 0 ,
'Subsoil' : 1})
test_lin_reg_bk_pol = test_lin_reg_bk_pol.drop(correlated_columns, axis=1)
```

```
row = the_funkcija(grid_lin_reg_polynomial, data_bez_korelacija, test_lin_reg_bk_pol,
"lin_reg_bez_korelacija_polynomial")
print(row)
metrics=pd.concat([metrics, pd.DataFrame([row])])
```

```
{'best_params_Ca': {'feature_generator': PolynomialFeatures(),
'model_lin_reg__fit_intercept': True, 'scaler': None}, 'best_params_P':
{'feature_generator': None, 'model_lin_reg__fit_intercept': False, 'scaler':
StandardScaler()}, 'best_params_pH': {'feature_generator': PolynomialFeatures(),
'model_lin_reg__fit_intercept': False, 'scaler': StandardScaler()}, 'best_params_SOC':
{'feature_generator': PolynomialFeatures(), 'model_lin_reg__fit_intercept': True,
'scaler': None}, 'best_params_Sand': {'feature_generator': PolynomialFeatures(),
'model_lin_reg__fit_intercept': False, 'scaler': None}, 'model':
'lin_reg_bez_korelacija_polynomial', 'Ca': 0.6420833025861913, 'P': 0.6988197190977719,
'pH': 0.6189733116872884, 'SOC': 0.766177253697109, 'Sand': 0.6706059297434439, 'MCRMSE':
0.6793319033623609}
```



```

pipeline_tree_polynomial = Pipeline([
    ('feature_generator', 'passthrough'),
    ('model_tree', model_tree),
])

parameters_tree_polynomial = {
    "model_tree__ccp_alpha" : [0.0, 0.05, 0.0025],
    "model_tree__max_depth" : [2, 5, 7, 9, 11],
    "model_tree__criterion" : ['squared_error', 'friedman_mse', 'absolute_error'],
    "feature_generator" : ['passthrough', PolynomialFeatures(degree=2)]
}

grid_tree_polynomial = GridSearchCV(pipeline_tree_polynomial, parameters_tree_polynomial,
n_jobs=-1, cv=5, scoring='neg_mean_squared_error')

```

```

test_tree_bk_pol = sorted_test.copy()
test_tree_bk_pol['Depth'] = test_tree_bk_pol['Depth'].replace({'Topsoil' : 0 , 'Subsoil'
: 1})
test_tree_bk_pol = test_tree_bk_pol.drop(correlated_columns, axis=1)

```

```

row = the_funkcija(grid_tree_polynomial, data_bez_korelacija, test_tree_bk_pol,
"tree_bez_korelacija_polynomial")
print(row)
metrics=pd.concat([metrics, pd.DataFrame([row])])

```

```

{'best_params_Ca': {'feature_generator': 'passthrough', 'model_tree__ccp_alpha': 0.0,
'feature_generator': 'passthrough', 'model_tree__ccp_alpha': 0.05,
'best_params_P': {'feature_generator': 'passthrough', 'model_tree__ccp_alpha': 0.0,
'best_params_pH': {'feature_generator': 'passthrough', 'model_tree__ccp_alpha': 0.0,
'best_params_SOC': {'feature_generator': 'passthrough', 'model_tree__ccp_alpha': 0.0,
'best_params_Sand': {'feature_generator': 'passthrough', 'model_tree__ccp_alpha': 0.0025,
'best_params_pH': 0.6163802595768648, 'SOC': 0.722614972588117, 'Sand': 0.5259174484953628, 'MCRMSE':
0.6201579821769898}

```

```

knn_pipeline_polynomial = Pipeline([
    ('scaler', StandardScaler()),
    ('feature_generator', PolynomialFeatures()),
    ("knn", knn),
])

parameters_knn_polynomial = {
    "knn__n_neighbors" : [3, 5, 7, 9],
    "knn__weights" : ['uniform', 'distance'],
    "knn__metric" : ['minkowski'],
    "knn__algorithm" : [ 'kd_tree', 'brute'],
    "feature_generator" : [None, PolynomialFeatures(degree=2)]
}

grid_knn_polynomial = GridSearchCV(knn_pipeline_polynomial, parameters_knn_polynomial,
n_jobs=-1 , cv=5, scoring="neg_mean_squared_error")

```

```

test_knn_bk_pol = sorted_test.copy()
test_knn_bk_pol['Depth'] = test_knn_bk_pol['Depth'].replace({'Topsoil' : 0 , 'Subsoil' : 1
})
test_knn_bk_pol = test_knn_bk_pol.drop(correlated_columns, axis=1)

```

```

row = the_funkcija(grid_knn_polynomial, data_bez_korelacija, test_knn_bk_pol,
"knn_bez_korelacija_polynomial")
print(row)
metrics=pd.concat([metrics, pd.DataFrame([row])])

{'best_params_Ca': {'feature_generator': None, 'knn__algorithm': 'kd_tree',
'knn__metric': 'minkowski', 'knn__n_neighbors': 7, 'knn__weights': 'distance'},
'best_params_P': {'feature_generator': None, 'knn__algorithm': 'kd_tree', 'knn__metric':
'minkowski', 'knn__n_neighbors': 7, 'knn__weights': 'uniform'}, 'best_params_pH':
{'feature_generator': None, 'knn__algorithm': 'kd_tree', 'knn__metric': 'minkowski',
'knn__n_neighbors': 9, 'knn__weights': 'distance'}, 'best_params_SOC':
{'feature_generator': None, 'knn__algorithm': 'kd_tree', 'knn__metric': 'minkowski',
'knn__n_neighbors': 9, 'knn__weights': 'distance'}, 'best_params_Sand':
{'feature_generator': None, 'knn__algorithm': 'kd_tree', 'knn__metric': 'minkowski',
'knn__n_neighbors': 5, 'knn__weights': 'distance'}, 'model':
'knn_bez_korelacija_polynomial', 'Ca': 0.6794565588462201, 'P': 0.7635625755021704, 'pH':
0.56808776830409, 'SOC': 0.6291466105575492, 'Sand': 0.543697402020544, 'MCRMSE':
0.6367901830461147}

```

```

random_forest_pipeline_polynomial = Pipeline([
    ('feature_generator', PolynomialFeatures()),
    ("random_forest", forest),
])

parameters_random_forest_polynomial = {
    "random_forest__n_estimators" : [50, 100, 150, 200],
    "random_forest__criterion" : ['squared_error', 'absolute_error', 'friedman_mse'],
    "random_forest__max_depth" : [2, 5, 7, 9],
    "feature_generator" : [None, PolynomialFeatures(degree=2)]
}

grid_random_forest_polynomial = GridSearchCV(random_forest_pipeline_polynomial,
parameters_random_forest_polynomial, n_jobs=-1 , cv=5, scoring="neg_mean_squared_error")

```

```

test_rf_bk_pol = sorted_test.copy()
test_rf_bk_pol['Depth'] = test_rf_bk_pol['Depth'].replace({'Topsoil' : 0 , 'Subsoil' : 1})
test_rf_bk_pol = test_rf_bk_pol.drop(correlated_columns, axis=1)

```

```

row = the_funkcija(grid_random_forest_polynomial, data_bez_korelacija, test_rf_bk_pol,
"random_forest_bez_korelacija_polynomial")
print(row)
metrics=pd.concat([metrics, pd.DataFrame([row])])

```

```

{'best_params_Ca': {'feature_generator': PolynomialFeatures(),
'random_forest__criterion': 'absolute_error', 'random_forest__max_depth': 9,
'random_forest__n_estimators': 100}, 'best_params_P': {'feature_generator': None,
'random_forest__criterion': 'friedman_mse', 'random_forest__max_depth': 9,
'random_forest__n_estimators': 100}, 'best_params_pH': {'feature_generator':
PolynomialFeatures(), 'random_forest__criterion': 'squared_error',
'random_forest__max_depth': 9, 'random_forest__n_estimators': 150}, 'best_params_SOC':
{'feature_generator': None, 'random_forest__criterion': 'friedman_mse',
'random_forest__max_depth': 9, 'random_forest__n_estimators': 200}, 'best_params_Sand':
{'feature_generator': None, 'random_forest__criterion': 'squared_error',
'random_forest__max_depth': 9, 'random_forest__n_estimators': 200}, 'model':
'random_forest_bez_korelacija_polynomial', 'Ca': 0.5110127894759193, 'P':
0.916817413665758, 'pH': 0.5015534832478835, 'SOC': 0.5635807763666445, 'Sand':
0.4158437032948581, 'MCRMSE': 0.5817616332102127}

```

Selekcija

```

from sklearn.feature_selection import SequentialFeatureSelector
selector = SequentialFeatureSelector(model_lin_reg, n_features_to_select=6, direction=
'forward')
pipeline_lin_reg_selection = Pipeline([
    ('scaler', StandardScaler()),
    ('feature_selector', selector),
    ('model_lin_reg', model_lin_reg),
])
lin_reg_params_selection = {
    'scaler' : [None, StandardScaler()],
    "feature_selector__direction" : ['forward', 'backward'],
    'model_lin_reg__fit_intercept' : [False, True],
}

grid_lin_reg_selection = GridSearchCV(pipeline_lin_reg_selection,
lin_reg_params_selection, n_jobs=-1, cv=5, scoring='neg_mean_squared_error')

```

```

test_lin_reg_bk_sel = sorted_test.copy()
test_lin_reg_bk_sel['Depth'] = test_lin_reg_bk_sel['Depth'].replace({'Topsoil' : 0 ,
'Subsoil' : 1})
test_lin_reg_bk_sel = test_lin_reg_bk_sel.drop(correlated_columns, axis=1)

```

```

row = the_funkcija(grid_lin_reg_selection, data_bez_korelacija, test_lin_reg_bk_sel,
"lin_reg_bez_korelacija_selection")
print(row)
metrics=pd.concat([metrics, pd.DataFrame([row])])

```

```

{'best_params_Ca': {'feature_selector__direction': 'backward',
'model_lin_reg__fit_intercept': False, 'scaler': None}, 'best_params_P':
{'feature_selector__direction': 'backward', 'model_lin_reg__fit_intercept': False,
'scaler': StandardScaler()}, 'best_params_pH': {'feature_selector__direction':
'backward', 'model_lin_reg__fit_intercept': False, 'scaler': StandardScaler()},
'best_params_SOC': {'feature_selector__direction': 'backward',
'model_lin_reg__fit_intercept': True, 'scaler': None}, 'best_params_Sand':
{'feature_selector__direction': 'backward', 'model_lin_reg__fit_intercept': False,
'scaler': StandardScaler()}, 'model': 'lin_reg_bez_korelacija_selection', 'Ca':
0.8283647579786129, 'P': 0.6920284204098044, 'pH': 0.6822460263000224, 'SOC':
0.8581477133798922, 'Sand': 0.8011313451718595, 'MCRMSE': 0.7723836526480382}

```

```

selector = SequentialFeatureSelector(knn, n_features_to_select=6, direction='forward')
knn_pipeline_selection = Pipeline([
    ('scaler', StandardScaler()),
    ('feature_selector', selector),
    ("knn", knn),
])

parameters_knn_selection = {
    "knn__n_neighbors" : [3, 5, 7, 9],
    "knn__weights" : ['uniform', 'distance'],
    "knn__metric" : ['minkowski'],
    "knn__algorithm" : [ 'kd_tree', 'brute'],
    "feature_selector__direction" : ['forward', 'backward'],
}

grid_knn_selection = GridSearchCV(knn_pipeline_selection, parameters_knn_selection,
n_jobs=-1 , cv=5, scoring="neg_mean_squared_error")

```

```

test_knn_bk_sel = sorted_test.copy()
test_knn_bk_sel['Depth'] = test_knn_bk_sel['Depth'].replace({'Topsoil' : 0 , 'Subsoil' : 1
})
test_knn_bk_sel = test_knn_bk_sel.drop(correlated_columns, axis=1)

```

```

row = the_funkcija(grid_knn_selection, data_bez_korelacija, test_knn_bk_sel,
"knn_bez_korelacija_selection")
print(row)
metrics=pd.concat([metrics, pd.DataFrame([row])])

```

```

{'best_params_Ca': {'feature_selector__direction': 'backward', 'knn__algorithm': 'brute',
'knn__metric': 'minkowski', 'knn__n_neighbors': 9, 'knn__weights': 'distance'},
'best_params_P': {'feature_selector__direction': 'forward', 'knn__algorithm': 'kd_tree',
'knn__metric': 'minkowski', 'knn__n_neighbors': 9, 'knn__weights': 'distance'},
'best_params_pH': {'feature_selector__direction': 'forward', 'knn__algorithm': 'brute',
'knn__metric': 'minkowski', 'knn__n_neighbors': 7, 'knn__weights': 'distance'},
'best_params_SOC': {'feature_selector__direction': 'backward', 'knn__algorithm': 'brute',
'knn__metric': 'minkowski', 'knn__n_neighbors': 9, 'knn__weights': 'distance'},
'best_params_Sand': {'feature_selector__direction': 'forward', 'knn__algorithm':
'kd_tree', 'knn__metric': 'minkowski', 'knn__n_neighbors': 9, 'knn__weights':
'distance'}, 'model': 'knn_bez_korelacija_selection', 'Ca': 0.5816067254419746, 'P':
0.6865540026830881, 'pH': 0.5371188972767886, 'SOC': 0.6045728658702846, 'Sand':
0.41334396574408644, 'MCRMSE': 0.5646392914032444}

```

```

pom = metrics.iloc[:,0:5]
metrics.drop(pom, inplace=True, axis=1)
metrics = pd.concat([metrics, pom])

```

```

metrics.to_csv("metrics.csv", index=False)

```

Multi Output regresija

- MultiOutputRegressor je metoda koji se bavi sa datasetovima koji imaju vise izlaznih numerickih varijabli
- Proverava da li postoji izmedju izlaznih varijabli korelecija koja moze da utice ne moguće predikcije

```
metrics_multiple = pd.DataFrame(columns=["model", "MCRMSE"])
```

```
from sklearn.multioutput import MultiOutputRegressor
from sklearn.metrics import mean_squared_error
#data.info()
def druga_funkcija(grid, X, Y, validacija_set, output_name):

    y_cols = list(Y.columns)

    X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, random_state=
7)

    grid.fit(X_train, Y_train)
    best_mo = grid.best_estimator_

    Y_pred = best_mo.predict(X_test)

    rmse_per_output = []
    for i in range(Y_pred.shape[1]):
        rmse_per_output.append(
            mean_squared_error(Y_test.iloc[:,i], Y_pred[:,i]) ** 0.5
        )

    metrics_multiple = {"model": output_name, "MCRMSE": float(np.mean(rmse_per_output))}

    PIDN = validacija_set["PIDN"].copy()
    X_valid = validacija_set.drop(columns=["PIDN"])

    # uskladi kolone kao na X (fit vreme)
    X_valid = X_valid.reindex(columns=X.columns, fill_value=0)

    pred_valid = best_mo.predict(X_valid)
    prediction = pd.DataFrame(pred_valid, columns=y_cols)
    prediction.insert(0, "PIDN", PIDN)

    prediction.to_csv(f"predictions/{output_name}.csv", index=False)
    return metrics_multiple
```

MODELI ZA MULTIPLE OUTPUT

```

multi_knn = MultiOutputRegressor(knn_pipeline)

parameters_knn_multi = {
    "estimator__knn__n_neighbors": [3, 5, 7, 9],
    "estimator__knn__weights": ['uniform', 'distance'],
    "estimator__knn__metric": ['minkowski'],
    "estimator__knn__algorithm": ['kd_tree', 'brute']
}

grid_knn_multi = GridSearchCV(multi_knn, parameters_knn_multi, n_jobs=-1, cv=5, scoring=
    "neg_mean_squared_error")

```

```

multi_forest = MultiOutputRegressor(random_forest_pipeline)

parameters_random_forest_multi = {
    "estimator__random_forest__n_estimators": [3, 5, 7],
    "estimator__random_forest__criterion": ['squared_error', 'absolute_error',
    'friedman_mse'],
    "estimator__random_forest__max_depth": [3, 5, 7]
}

grid_random_forest_multi = GridSearchCV(multi_forest, parameters_random_forest_multi,
    n_jobs=-1, cv=5, scoring="neg_mean_squared_error")

```

```

multi_lin_reg = MultiOutputRegressor(pipeline_lin_reg)

parameters_lin_reg_multi = {
    'estimator__scaler' : [None, StandardScaler()],
    'estimator__model__lin_reg__fit_intercept' : [False, True],
}

grid_lin_reg_multi = GridSearchCV(multi_lin_reg, parameters_lin_reg_multi, n_jobs=-1, cv=5
    , scoring='neg_mean_squared_error')

```

```

multi_tree = MultiOutputRegressor(pipeline_tree)

parameters_tree_multi = {
    "estimator__model__tree__ccp_alpha" : [0.0, 0.05, 0.0025],
    "estimator__model__tree__max_depth" : [2, 5, 7, 9, 11],
    "estimator__model__tree__criterion" : ['squared_error', 'friedman_mse',
    'absolute_error'],
}

grid_tree_multi = GridSearchCV(multi_tree, parameters_tree_multi, n_jobs=-1, cv=5,
    scoring='neg_mean_squared_error')

```

```
xg_boost_pipeline_multi = Pipeline ([
    ("model_xg_boost", MultiOutputRegressor(xgb.XGBRegressor(eval_metric='rmse',device=
'cpu'))))
])

grid_xg_boost_multi = GridSearchCV(
    xg_boost_pipeline_multi,
    {
        "model_xg_boost__estimator__max_depth": [4, 5, 6],
        "model_xg_boost__estimator__n_estimators": [160, 180, 200],
        "model_xg_boost__estimator__learning_rate": [0.01, 0.015],
    },
    n_jobs=-1,
    cv=5,
    scoring="neg_mean_squared_error"
)
```

Implementacija grid tj. modela nad gridom

KNN

- bez icega
- clipped
- bez korelacija

```
test_knn_multiple_output = sorted_test.copy()
test_knn_multiple_output['Depth'] = test_knn_multiple_output['Depth'].replace({'Topsoil'
: 0 , 'Subsoil' : 1})
test_knn_multiple_output = test_knn_multiple_output.drop(correlated_columns, axis=1)
```

```
row_multi = druga_funkcija(grid_knn_multi,X, Y, test_knn_multiple_output, "knn_multi")
print(row_multi)
metrics_multiple=pd.concat([metrics_multiple, pd.DataFrame([row_multi])])
```

```
{'model': 'knn_multi', 'MCRMSE': 0.5330502307031515}
```

```
test_knn_multiple_output_clipped = sorted_test.copy()
test_knn_multiple_output_clipped['Depth'] = test_knn_multiple_output_clipped['Depth'].
replace({'Topsoil' : 0 , 'Subsoil' : 1})
test_knn_multiple_output_clipped = test_knn_multiple_output_clipped.drop
(correlated_columns, axis=1)
```



```

row_multi = druga_funkcija(grid_knn_multi,X_clipped, Y_clipped,
test_knn_multiple_output_clipped, "knn_multi_clipped")
print(row_multi)
metrics_multiple=pd.concat([metrics_multiple, pd.DataFrame([row_multi])])

{'model': 'knn_multi_clipped', 'MCRMSE': 0.5320347873605555}

```

```

test_knn_multiple_output_bk = sorted_test.copy()
test_knn_multiple_output_bk['Depth'] = test_knn_multiple_output_bk['Depth'].replace({
'Topsoil' : 0 , 'Subsoil' : 1})
test_knn_multiple_output_bk = test_knn_multiple_output_bk.drop(correlated_columns, axis=1)

```

```

row_multi = druga_funkcija(grid_knn_multi,X_bez_korelacija, Y_bez_korelacija,
test_knn_multiple_output_bk, "knn_multi_bez_korelacija")
print(row_multi)
metrics_multiple=pd.concat([metrics_multiple, pd.DataFrame([row_multi])])

{'model': 'knn_multi_bez_korelacija', 'MCRMSE': 0.6488063475048892}

```

Random Forest pokusaj

- bez icega
- clipped
- bez korelacija

```

test_random_forest_multiple_output = sorted_test.copy()
test_random_forest_multiple_output['Depth'] = test_random_forest_multiple_output['Depth'].
replace({'Topsoil' : 0 , 'Subsoil' : 1})
test_random_forest_multiple_output = test_random_forest_multiple_output.drop
(correlated_columns, axis=1)

```

```

row_multi = druga_funkcija(grid_random_forest_multi, X, Y,
test_random_forest_multiple_output, "random_forest_multi")
print(row_multi)
metrics_multiple=pd.concat([metrics_multiple, pd.DataFrame([row_multi])])

{'model': 'random_forest_multi', 'MCRMSE': 0.5687284557658259}

```

```

test_random_forest_multiple_output_clipped = sorted_test.copy()
test_random_forest_multiple_output_clipped['Depth'] =
test_random_forest_multiple_output_clipped['Depth'].replace({'Topsoil' : 0 , 'Subsoil' : 1
})
test_random_forest_multiple_output_clipped = test_random_forest_multiple_output_clipped.
drop(correlated_columns, axis=1)

```

```
row_multi = druga_funkcija(grid_random_forest_multi, X_clipped, Y_clipped,
test_random_forest_multiple_output_clipped, "random_forest_multi_clipped")
print(row_multi)
metrics_multiple=pd.concat([metrics_multiple, pd.DataFrame([row_multi])])

{'model': 'random_forest_multi_clipped', 'MCRMSE': 0.5594127728521754}
```

```
test_random_forest_multiple_output_bk = sorted_test.copy()
test_random_forest_multiple_output_bk['Depth'] = test_random_forest_multiple_output_bk[
'Depth'].replace({'Topsoil' : 0 , 'Subsoil' : 1})
test_random_forest_multiple_output_bk = test_random_forest_multiple_output_bk.drop
(correlated_columns, axis=1)
```

```
row_multi = druga_funkcija(grid_random_forest_multi, X_bez_korelacija, Y_bez_korelacija,
test_random_forest_multiple_output_clipped, "random_forest_multi_bez_korelacija")
print(row_multi)
metrics_multiple=pd.concat([metrics_multiple, pd.DataFrame([row_multi])])

{'model': 'random_forest_multi_bez_korelacija', 'MCRMSE': 0.6139218852802328}
```

Linearna regresija

- bez icega
- clipped
- bez korelacija

```
test_lin_reg_multiple_output = sorted_test.copy()
test_lin_reg_multiple_output['Depth'] = test_lin_reg_multiple_output['Depth'].replace({
'Topsoil' : 0 , 'Subsoil' : 1})
test_lin_reg_multiple_output = test_lin_reg_multiple_output.drop(correlated_columns, axis=
1)
```

```
row_multi = druga_funkcija(grid_lin_reg_multi, X , Y , test_lin_reg_multiple_output,
"lin_reg_multi")
print(row_multi)
metrics_multiple=pd.concat([metrics_multiple, pd.DataFrame([row_multi])])

{'model': 'lin_reg_multi', 'MCRMSE': 0.723304090844857}
```

```
test_lin_reg_multiple_output_clipped = sorted_test.copy()
test_lin_reg_multiple_output_clipped['Depth'] = test_lin_reg_multiple_output_clipped[
'Depth'].replace({'Topsoil' : 0 , 'Subsoil' : 1})
test_lin_reg_multiple_output_clipped = test_lin_reg_multiple_output_clipped.drop
(correlated_columns, axis=1)
```

```
row_multi = druga_funkcija(grid_lin_reg_multi, X_clipped , Y_clipped ,
test_lin_reg_multiple_output_clipped, "lin_reg_multi_clipped")
print(row_multi)
metrics_multiple=pd.concat([metrics_multiple, pd.DataFrame([row_multi])])

{'model': 'lin_reg_multi_clipped', 'MCRMSE': 0.862346800344957}
```

```
test_lin_reg_multiple_output_bk = sorted_test.copy()
test_lin_reg_multiple_output_bk['Depth'] = test_lin_reg_multiple_output_bk['Depth'].
replace({'Topsoil' : 0 , 'Subsoil' : 1})
test_lin_reg_multiple_output_bk = test_lin_reg_multiple_output_bk.drop
(correlated_columns, axis=1)
```

```
row_multi = druga_funkcija(grid_lin_reg_multi, X_bez_korelacija , Y_bez_korelacija ,
test_lin_reg_multiple_output_bk, "lin_reg_multi_bez_korelacija")
print(row_multi)
metrics_multiple=pd.concat([metrics_multiple, pd.DataFrame([row_multi])])

{'model': 'lin_reg_multi_bez_korelacija', 'MCRMSE': 0.7628467874576359}
```

Regresiono stablo

- bez icega
- clipped
- bez korelacija

```
test_tree_multiple_output = sorted_test.copy()
test_tree_multiple_output['Depth'] = test_tree_multiple_output['Depth'].replace({'Topsoil'
: 0 , 'Subsoil' : 1})
test_tree_multiple_output = test_tree_multiple_output.drop(correlated_columns, axis=1)
```

```
row_multi = druga_funkcija(grid_tree_multi,X, Y, test_tree_multiple_output, "tree_multi")
print(row_multi)
metrics_multiple=pd.concat([metrics_multiple, pd.DataFrame([row_multi])])

{'model': 'tree_multi', 'MCRMSE': 0.7212997299228847}
```

```
test_tree_multiple_output_clipped = sorted_test.copy()
test_tree_multiple_output_clipped['Depth'] = test_tree_multiple_output_clipped['Depth'].
replace({'Topsoil' : 0 , 'Subsoil' : 1})
test_tree_multiple_output_clipped = test_tree_multiple_output_clipped.drop
(correlated_columns, axis=1)
```

```
row_multi = druga_funkcija(grid_tree_multi, X_clipped , Y_clipped ,
test_tree_multiple_output_clipped, "tree_multi_clipped")
print(row_multi)
metrics_multiple=pd.concat([metrics_multiple, pd.DataFrame([row_multi])])

{'model': 'tree_multi_clipped', 'MCRMSE': 0.7266971161233039}
```

```
test_tree_multiple_output_bk = sorted_test.copy()
test_tree_multiple_output_bk['Depth'] = test_tree_multiple_output_bk['Depth'].replace({
'Topsoil' : 0 , 'Subsoil' : 1})
test_tree_multiple_output_bk = test_tree_multiple_output_bk.drop(correlated_columns, axis=
1)
```

```
row_multi = druga_funkcija(grid_tree_multi, X_bez_korelacija , Y_bez_korelacija ,
test_tree_multiple_output, "tree_multi_bez_korelacija")
print(row_multi)
metrics_multiple=pd.concat([metrics_multiple, pd.DataFrame([row_multi])])

{'model': 'tree_multi_bez_korelacija', 'MCRMSE': 0.7106908706764448}
```

XGboost

- bez korelacija

```
test_xg_boost_multiple_output_bk = sorted_test.copy()
test_xg_boost_multiple_output_bk['Depth'] = test_xg_boost_multiple_output_bk['Depth'].
replace({'Topsoil' : 0 , 'Subsoil' : 1})
test_xg_boost_multiple_output_bk = test_xg_boost_multiple_output_bk.drop
(correlated_columns, axis=1)
```

```

X_bez_korelacija["Depth"] = pd.to_numeric(X_bez_korelacija["Depth"], errors="coerce")
test_xg_boost_multiple_output_bk["Depth"] = pd.to_numeric(test_xg_boost_multiple_output_bk
["Depth"], errors="coerce")

# print(X.dtypes)
# print(Y.dtypes)
# print(test_xg_boost_multiple_output_bk.dtypes)

valid_copy_bk_multi = test_xg_boost_multiple_output_bk.copy()
valid_copy_bk_multi["PIDN"] = sorted_test["PIDN"]

row_multi = druga_funkcija(grid_xg_boost_multi, X_bez_korelacija , Y_bez_korelacija ,
test_xg_boost_multiple_output_bk, "xg_boost_multi_bez_korelacija")
print(row_multi)
metrics_multiple=pd.concat([metrics_multiple, pd.DataFrame([row_multi])])

# print(sorted_test.columns)
# print(test_xg_boost_bk.columns)

{'model': 'xg_boost_multi_bez_korelacija', 'MCRMSE': 0.6494658495527119}

```

```

metrics_multiple.to_csv("metrics_multiple.csv", index=False)

```

metrics_multiple

	model	MCRMSE
0	knn_multi	0.53305
0	knn_multi_clipped	0.532035
0	knn_multi_bez_korelacija	0.648806
0	random_forest_multi	0.568728
0	random_forest_multi_clipped	0.559413
0	random_forest_multi_bez_korelacija	0.613922
0	lin_reg_multi	0.723304
0	lin_reg_multi_clipped	0.862347
0	lin_reg_multi_bez_korelacija	0.762847
0	tree_multi	0.7213
0	tree_multi_clipped	0.726697
0	tree_multi_bez_korelacija	0.710691
0	xg_boost_multi_bez_korelacija	0.649466

Regressor chain

Modeli za Regressor chain

```
metrics_reg_chain = pd.DataFrame(columns=["model", "MCRMSE"])
```

```
from sklearn.multioutput import RegressorChain
reg_chain_knn = RegressorChain(knn_pipeline, order = [1, 2, 3, 0, 4])

parameters_knn_reg_chain = {
    "estimator__knn__n_neighbors": [3, 5, 7, 9],
    "estimator__knn__weights": ['uniform', 'distance'],
    "estimator__knn__metric": ['minkowski'],
    "estimator__knn__algorithm": ['kd_tree', 'brute']
}

grid_knn_reg_chain = GridSearchCV(reg_chain_knn, parameters_knn_reg_chain, n_jobs=-1, cv=5,
    scoring="neg_mean_squared_error")
```

```
reg_chain_lin_reg = RegressorChain(pipeline_lin_reg, order = [1, 2, 3, 0, 4])

parameters_lin_reg_reg_chain = {
    'estimator__scaler' : [None, StandardScaler()],
    'estimator__model__lin_reg__fit_intercept' : [False, True],
}

grid_lin_reg_reg_chain = GridSearchCV(reg_chain_lin_reg, parameters_lin_reg_reg_chain,
    n_jobs=-1, cv=5, scoring='neg_mean_squared_error')
```

```
reg_chain_tree = RegressorChain(pipeline_tree, order = [0, 1, 2, 3, 4])

parameters_tree_reg_chain = {
    "estimator__model__tree__ccp_alpha" : [0.0, 0.05, 0.0025],
    "estimator__model__tree__max_depth" : [2, 5, 7, 9, 11],
    "estimator__model__tree__criterion" : ['squared_error', 'friedman_mse',
    'absolute_error'],
}

grid_tree_reg_chain = GridSearchCV(reg_chain_tree, parameters_tree_reg_chain, n_jobs=-1,
    cv=5, scoring='neg_mean_squared_error')
```

Izvrševanje Regressor chain nad modelima

KNN

- bez icega
- clipped
- bez korelacija

```
test_knn_reg_chain = sorted_test.copy()
test_knn_reg_chain['Depth'] = test_knn_reg_chain['Depth'].replace({'Topsoil' : 0 ,
'Subsoil' : 1})
test_knn_reg_chain = test_knn_reg_chain.drop(correlated_columns, axis=1)
```

```
row_reg_chain = druga_funkcija(grid_knn_reg_chain,X, Y, test_knn_reg_chain,
"knn_regressor_chain")
print(row_reg_chain)
metrics_reg_chain=pd.concat([metrics_reg_chain, pd.DataFrame([row_reg_chain])])
```

```
{'model': 'knn_regressor_chain', 'MCRMSE': 0.5327570222969148}
```

```
test_knn_reg_chain_clipped = sorted_test.copy()
test_knn_reg_chain_clipped['Depth'] = test_knn_reg_chain_clipped['Depth'].replace({'Topsoil' : 0 , 'Subsoil' : 1})
test_knn_reg_chain_clipped = test_knn_reg_chain_clipped.drop(correlated_columns, axis=1)
```

```
row_reg_chain = druga_funkcija(grid_knn_reg_chain,X_clipped, Y_clipped,
test_knn_reg_chain_clipped, "knn_regressor_chain_clipped")
print(row_reg_chain)
metrics_reg_chain=pd.concat([metrics_reg_chain, pd.DataFrame([row_reg_chain])])
```

```
{'model': 'knn_regressor_chain_clipped', 'MCRMSE': 0.5317798669688398}
```

```
test_knn_reg_chain_bk = sorted_test.copy()
test_knn_reg_chain_bk['Depth'] = test_knn_reg_chain_bk['Depth'].replace({'Topsoil' : 0 ,
'Subsoil' : 1})
test_knn_reg_chain_bk = test_knn_reg_chain_bk.drop(correlated_columns, axis=1)
```

```
row_reg_chain = druga_funkcija(grid_knn_reg_chain,X_bez_korelacija, Y_bez_korelacija,
test_knn_multiple_output_bk, "knn_regressor_chain_bez_korelacija")
print(row_reg_chain)
metrics_reg_chain=pd.concat([metrics_reg_chain, pd.DataFrame([row_reg_chain])])
```

```
{'model': 'knn_regressor_chain_bez_korelacija', 'MCRMSE': 0.6551982066307646}
```

Linearna regresija

- bez icega
- clipped
- bez korelacija

```
test_lin_reg_reg_chain = sorted_test.copy()
test_lin_reg_reg_chain['Depth'] = test_lin_reg_reg_chain['Depth'].replace({'Topsoil' : 0
, 'Subsoil' : 1})
test_lin_reg_reg_chain = test_lin_reg_reg_chain.drop(correlated_columns, axis=1)
```

```
row_reg_chain = druga_funkcija(grid_lin_reg_reg_chain, X , Y , test_lin_reg_reg_chain,
"lin_reg_regressor_chain")
print(row_reg_chain)
metrics_reg_chain=pd.concat([metrics_reg_chain, pd.DataFrame([row_reg_chain])])
```

```
{'model': 'lin_reg_regressor_chain', 'MCRMSE': 0.723304090845095}
```

```
test_lin_reg_reg_chain_clipped = sorted_test.copy()
test_lin_reg_reg_chain_clipped['Depth'] = test_lin_reg_reg_chain_clipped['Depth'].replace
({'Topsoil' : 0 , 'Subsoil' : 1})
test_lin_reg_reg_chain_clipped = test_lin_reg_reg_chain_clipped.drop(correlated_columns,
axis=1)
```

```
row_reg_chain = druga_funkcija(grid_lin_reg_reg_chain, X_clipped , Y_clipped ,
test_lin_reg_reg_chain_clipped, "lin_reg_regressor_chain_clipped")
print(row_reg_chain)
metrics_reg_chain=pd.concat([metrics_reg_chain, pd.DataFrame([row_reg_chain])])
```

```
{'model': 'lin_reg_regressor_chain_clipped', 'MCRMSE': 0.8623468003441825}
```

```
test_lin_reg_reg_chain_bk = sorted_test.copy()
test_lin_reg_reg_chain_bk['Depth'] = test_lin_reg_reg_chain_bk['Depth'].replace({'Topsoil'
: 0 , 'Subsoil' : 1})
test_lin_reg_reg_chain_bk = test_lin_reg_reg_chain_bk.drop(correlated_columns, axis=1)
```

```
row_reg_chain = druga_funkcija(grid_lin_reg_reg_chain, X_bez_korelacija ,
Y_bez_korelacija , test_lin_reg_reg_chain_bk, "lin_reg_regressor_chain_bez_korelacija")
print(row_reg_chain)
metrics_reg_chain=pd.concat([metrics_reg_chain, pd.DataFrame([row_reg_chain])])
```

```
{'model': 'lin_reg_regressor_chain_bez_korelacija', 'MCRMSE': 0.7628467874576357}
```


Regresiono stablo

- bez icega
- clipped
- bez korelacija

```
test_tree_reg_chain = sorted_test.copy()
test_tree_reg_chain['Depth'] = test_tree_reg_chain['Depth'].replace({'Topsoil' : 0 ,
'Subsoil' : 1})
test_tree_reg_chain = test_tree_reg_chain.drop(correlated_columns, axis=1)
```

```
row_reg_chain = druga_funkcija(grid_tree_reg_chain,X, Y, test_tree_reg_chain,
"tree_regressor_chain")
print(row_reg_chain)
metrics_reg_chain=pd.concat([metrics_reg_chain, pd.DataFrame([row_reg_chain])])

{'model': 'tree_regressor_chain', 'MCRMSE': 0.6758993354985978}
```

```
test_tree_reg_chain_clipped = sorted_test.copy()
test_tree_reg_chain_clipped['Depth'] = test_tree_reg_chain_clipped['Depth'].replace({'
Topsoil' : 0 , 'Subsoil' : 1})
test_tree_reg_chain_clipped = test_tree_reg_chain_clipped.drop(correlated_columns, axis=1)
```

```
row_reg_chain = druga_funkcija(grid_tree_reg_chain, X_clipped , Y_clipped ,
test_tree_reg_chain_clipped, "tree_regressor_chain_clipped")
print(row_reg_chain)
metrics_reg_chain=pd.concat([metrics_reg_chain, pd.DataFrame([row_reg_chain])])

{'model': 'tree_regressor_chain_clipped', 'MCRMSE': 0.6549045856677145}
```

```
test_tree_reg_chain_bk = sorted_test.copy()
test_tree_reg_chain_bk['Depth'] = test_tree_reg_chain_bk['Depth'].replace({'Topsoil' : 0
, 'Subsoil' : 1})
test_tree_reg_chain_bk = test_tree_reg_chain_bk.drop(correlated_columns, axis=1)
```

```
row_reg_chain = druga_funkcija(grid_tree_reg_chain, X_bez_korelacija , Y_bez_korelacija ,
test_tree_reg_chain_bk, "tree_regressor_chain_bez_korelacija")
print(row_reg_chain)
metrics_reg_chain=pd.concat([metrics_reg_chain, pd.DataFrame([row_reg_chain])])

{'model': 'tree_regressor_chain_bez_korelacija', 'MCRMSE': 0.7683701421127218}
```

metrics_reg_chain

	model	MCRMSE
0	knn_regressor_chain	0.532757
0	knn_regressor_chain_clipped	0.53178
0	knn_regressor_chain_bez_korelacija	0.655198
0	lin_reg_regressor_chain	0.723304
0	lin_reg_regressor_chain_clipped	0.862347
0	lin_reg_regressor_chain_bez_korelacija	0.762847
0	tree_regressor_chain	0.675899
0	tree_regressor_chain_clipped	0.654905
0	tree_regressor_chain_bez_korelacija	0.76837

```
metrics_reg_chain.to_csv("metrics_reg_chain.csv", index=False)
```